METRIC

A-A-59231C
16 July 2018
SUPERSEDING
A-A-59231B
9 July 2012

COMMERCIAL ITEM DESCRIPTION

ETHANOLAMINES (MONOETHANOLAMINE AND TRIETHANOLAMINE), TECHNICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers technical grades of commercial monoethanolamine and triethanolamine. Monoethanolamine is intended for use as a decontaminant for riot control agent CS (ortho-chlorobenzylidene malononitrile) and for the chemical agent CS1 dispersing equipment. Monoethanolamine is also intended for the removal of carbon dioxide from air and other gases. Triethanolamine is intended for use when an organic base is required, as in maintaining alkalinity of water solutions to prevent corrosion in such equipment as boilers, pipes, and radiators.

2. CLASSIFICATION. The ethanolamines shall be of the following types as specified (see 7.3(b)):

2.1 Type.

Type I - Monoethanolamine (MEA)
Type II - Triethanolamine (TEA)

3. SALIENT CHARACTERISTICS.

3.1 Monoethanolamine. MEA supplied under this CID shall meeting the following characteristics:

3.1.1 Appearance. MEA shall be clear, viscous, and free of suspended matter in the liquid state. When in the solid state, MEA shall have a white, crystalline appearance, without visual evidence of impurities or discoloration.

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3.1.2 Specific Gravity. Specific gravity at 20 ºC/ 20 ºC shall be within the range of 1.017 to 1.021.

3.1.3 Color. Color shall have a maximum of 15 on the platinum-cobalt scale, tested in accordance with ASTM D1209.

3.1.4 Water Content. The maximum allowable water content shall not exceed 0.30% by weight, test in accordance with ASTM E203.

3.1.5 Purity. Assay shall be conducted by gas chromatography and shall meet the purity specifications within Table I.

TABLE I. Purity Specification for MEA

Component	Purity Requirement
MEA (weight percent)	98.0 % (minimum)
TEA (weight percent)	1.0 % (maximum)
DEA (weight percent)	1.5 % (maximum)
Iron (weight percent)	0.003% (maximum)

3.2 Triethanolamine. TEA supplied under this CID shall meeting the following characteristics:

3.2.1 Appearance. TEA shall be clear, viscous, and free of suspended matter in the liquid state. When in the solid state, TEA shall have a white, crystalline appearance, without visual evidence of impurities or discoloration.

3.2.2 Specific Gravity. Specific gravity at 20 ºC/ 20 ºC shall be within the range of 1.124 to 1.129.

3.2.3 Color. Color shall have a maximum of 40 on the platinum-cobalt scale, tested in accordance with ASTM D1209.

3.2.4 Water Content. The maximum allowable water content shall not exceed 0.20% by weight, tested in accordance with ASTM E203.

3.2.5 Purity. Assay shall be conducted by gas chromatography and shall meet the purity specifications within Table II.

TABLE II. Purity Specification for TEA

Component	Purity Requirement
TEA(weight percent)	99.0 % (minimum)
MEA (weight percent)	0.50 % (maximum)
DEA (weight percent)	0.50 % (maximum)

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4. REGULATORY REQUIREMENTS.

4.1 Marking, packaging, and labeling. Material shall be labeled, packed, and marked in accordance with Title 49, Code of Federal Regulations (CFR) paragraphs 100 to 185.

4.2 Recycled materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.3 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 CFR 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

5. PRODUCT CONFORMANCE PROVISIONS.

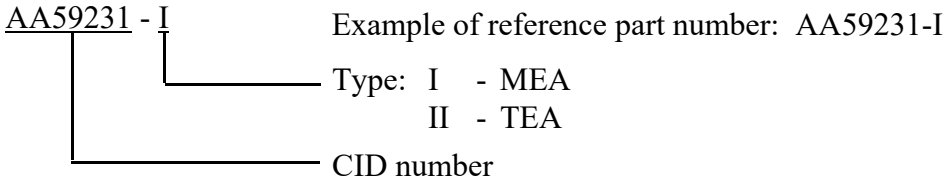
5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market and the offeror must be able to show test data or lab results of meeting the salient characteristics of the ethanolamines specified herein.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(c)).

7. NOTES.

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 CFR. CFR documents may be obtained from https://www.ecfr.gov/ .

7.2.2 FAR. A copy of FAR documents may be obtained from https://www.acquisition.gov/.

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7.2.3 Federal standards. Copies of federal standards may be obtained from http://quicksearch.dla.mil/.

7.2.4 ASTM International standards. Electronic copies of ASTM standards may be obtained from https://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type of ethanolamine required (see 2).
c. Packaging requirements (see 6).
d. Unit of issue and quantity.

7.4 National Stocking Numbers (NSNs). Table III provides a listing of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

TABLE III. NSNs Assigned to A-A-59231

NSN	A-A-59231 Type
6810-00-075-6876	I
6810-00-270-6207	I
6810-00-281-2192	II
6810-00-922-0866	I

7.5 Sampling and testing precautions. This CID requires inspection and use of chemical material which is potentially dangerous to personnel. All applicable safety rules, regulations, and procedures must be followed in the sampling and testing of this material.

7.6 Significant figures. For the purpose of determining conformance with this CID, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29, "Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications".

7.7 Subject term (key word) listing.

Chemical agent CS1
DEA
Diethanolamine
Gas chromatography
MEA
Riot control agent CS
TEA

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MILITARY INTERESTS:

Custodians:
Army – CR4
Navy – SH
Air Force – 68
DLA – GS

Review Activity:
Army – SM

CIVILIAN AGENCY
COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – GS3

(Project 6810-2017-010)

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